

Exp no:9**Monitoring and Logging Fabric Networks****Aim**

To monitor the Hyperledger Fabric network by inspecting Docker containers, viewing peer and orderer logs, checking network status, monitoring resource utilization, and analyzing blockchain transaction logs for troubleshooting and performance analysis

procedure:**Step 1: Navigate to the Fabric Test Network**

None

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

None

```
./network.sh up createChannel -ca
```

Step 3: Set the Fabric Environment

None

```
export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=$PWD/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizat
ions/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
```

```
export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations
/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.c
om/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.c
om-cert.pem
```

Step 4: Verify Running Docker Containers

None

```
docker ps
```

```

[ram@ram] ~/fabric-dev/fabric-samples/test-network
$ docker ps

```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
b146bdf4db29	hyperledger/fabric-peer:latest	peer0.org1.example.com	"peer node start"	About a minute ago	Up 59 seconds	0.0.0.0:7051→7051/tcp, [::]:7051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp
939bf681ac45	hyperledger/fabric-orderer:latest	orderer.example.com	"orderer"	About a minute ago	Up 59 seconds	0.0.0.0:7050→7050/tcp, [::]:7050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/t
13095844313a	hyperledger/fabric-peer:latest	peer0.org2.example.com	"peer node start"	About a minute ago	Up 59 seconds	0.0.0.0:9051→9051/tcp, [::]:9051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp
3749d8967919	hyperledger/fabric-ca:latest	ca_orderer	"sh -c 'fabric-ca-se..."	About a minute ago	Up About a minute	0.0.0.0:9054→9054/tcp, [::]:9054→9054/tcp, 7054/tcp, 0.0.0.0:19054→19054/tcp, [::]:19054→19054/tcp
c690c030ef9c	hyperledger/fabric-ca:latest	ca_org1	"sh -c 'fabric-ca-se..."	About a minute ago	Up About a minute	0.0.0.0:7054→7054/tcp, [::]:7054→7054/tcp, 0.0.0.0:17054→17054/tcp, [::]:17054→17054/tcp
8fa48dcde8cf	hyperledger/fabric-ca:latest	ca_org2	"sh -c 'fabric-ca-se..."	About a minute ago	Up About a minute	0.0.0.0:8054→8054/tcp, [::]:8054→8054/tcp, 7054/tcp, 0.0.0.0:18054→18054/tcp, [::]:18054→18054/tcp
ac23d975a119	ghcr.io/eramba/eramba:latest	cron	"/docker-cron-entryp..."	3 weeks ago	Up 4 hours	80/tcp
2e575645bc7e	ghcr.io/eramba/eramba-mcp-server:latest	mcp_server	"docker-entrypoint.s..."	3 weeks ago	Up 4 hours (healthy)	3000/tcp
fa96f6dc84bd	ghcr.io/eramba/eramba:latest	eramba	"/docker-entrypoint..."	3 weeks ago	Up 4 hours	80/tcp
03e10bcef0b0	redis:7.4.2-alpine	redis	"docker-entrypoint.s..."	3 weeks ago	Up 4 hours	
da98fac37466	mysql:8.4.3-oracle	mysql	"docker-entrypoint.s..."	3 weeks ago	Up 4 hours	3306/tcp, 33060/tcp

Step 5: Check Network Status

None

```
docker ps -a
```

Shows both running and stopped containers with their status.

7ecc4047511f	9494c7ebca73	admiring_colden	"docker-entrypoint.s..."	About an hour ago	Created
8c294fce18f5	da609efd28db	reverent_rhodes	"docker-entrypoint.s..."	3 hours ago	Created
73a4d266e4fc	a17c59f99444	jolly_ramanujan	"docker-entrypoint.s..."	3 hours ago	Created
cabc3795c581	fa9dbec4d603	awesome_jackson	"docker-entrypoint.s..."	3 hours ago	Created
5de6d8fd98a0	222762343836	vigilant_diffie	"docker-entrypoint.s..."	3 hours ago	Created
c528e9057fee	8ee81fb551d6	trusting_mahavira	"docker-entrypoint.s..."	4 hours ago	Created
cf4b1e5777ea	874a8d08e5c2	sweet_carson	"docker-entrypoint.s..."	4 hours ago	Created
5071e8f4773f	0fbe7039a5d2	exciting_colden	"docker-entrypoint.s..."	4 hours ago	Created
bc90a90e44e0	8be1e1881b25	wizardly_shamir	"docker-entrypoint.s..."	4 hours ago	Created
6f5296e9cf4f	fe36be5e1952	lucid_swanson	"docker-entrypoint.s..."	5 days ago	Created
f5560bb2bc98	7a396217cf40	unruffled_meninsky	"docker-entrypoint.s..."	5 days ago	Created
b5b0de6e6550	2ad82d91c40f	mystifying_kare	"docker-entrypoint.s..."	5 days ago	Created

Step 6: Deploy the Asset Transfer Chaincode

None

```
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript
```

Expected:

Packaging → Installing → Approving → Committing → Chaincode deployed successfully.

Step 7: Verify Installed Chaincode

None

```
peer lifecycle chaincode queryinstalled
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ peer lifecycle chaincode queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629, Label: basic_1.0
```

Step 8: Monitor Peer Logs

None

```
docker logs peer0.org1.example.com
```

For recent logs only:

None

```
docker logs --tail 50 peer0.org1.example.com
```

Expected log entries include peer startup, port 7051 listening, and chaincode installation.

Step 9: Monitor Orderer Logs

None

```
docker logs orderer.example.com
```

Recent logs only:

None

```
docker logs --tail 50 orderer.example.com
```

Expected log entries include orderer startup, channel initialization, and transaction processing.

Step 10: Verify the Channel

None

```
peer channel list
```

Expected:

None

Channels peers has joined:
mychannel